H.T.No: | | | | | | | | | | |  Course Code: **2501CS71**

# ADITYA UNIVERSITY

B.Tech – I Semester End Examinations Regular (AR24) – Nov 2024

## COMPUTER ORGANIZATION & ARCHITECTURE

(Common to CSE, CSE(DS), IT & AIML)

**Time: 3 hours** **Max. Marks: 100**

**Answer ONE question from each unit**
**All Questions Carry Equal Marks**
**All parts of the questions must be answered at one place only**

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| **UNIT-I** | | | | | |
| **1** | **a.** | Explain system bus structure and describe address bus, data bus and control bus. | L2 | CO1 | [10M] |
| | **b.** | Convert the following.<br>i) $(41.6875)_{10}$ to Binary<br>ii) $(736.4)_8$ to Binary<br>iii) $(7A3.F)_{16}$ to Binary<br>iv) $(101011)_2$ to Gray<br>v) $(47)_{10}$ to Excess-3 | L3 | CO1 | [10M] |
| | | **(OR)** | | | |
| **2** | **a.** | Illustrate the 2's complement addition and subtraction method with a neat flowchart. | L2 | CO1 | [10M] |
| | **b.** | Multiply the following using Booth algorithm: **(+5) × (−4)** | L3 | CO1 | [10M] |
| **UNIT-II** | | | | | |
| **3** | **a.** | Construct the common bus system using three-state buffer with help of neat diagram. Explain how register selection is performed in the bus system. | L3 | CO2 | [10M] |
| | **b.** | Describe logic microoperations. Explain the basic logic microoperations with examples. | L2 | CO2 | [10M] |
| | | **(OR)** | | | |
| **4** | **a.** | List the computer registers of a basic computer and state the function of each register. | L1 | CO2 | [10M] |
| | **b.** | Infer the interrupt cycle and explain how an interrupt is handled in the basic computer. | L2 | CO2 | [10M] |
| **UNIT-III** | | | | | |
| **5** | **a.** | Explain the general register organization of a CPU with a neat block diagram. | L2 | CO3 | [10M] |
| | **b.** | Illustrate the data transfer instructions with examples. | L2 | CO3 | [10M] |
| | | **(OR)** | | | |
| **6** | **a.** | Discuss the Microprogrammed control organization in control memory. | L2 | CO3 | [10M] |
| | **b.** | Explain microinstruction format with suitable examples. | L2 | CO3 | [10M] |
| **UNIT-IV** | | | | | |
| **7** | **a.** | Explain the memory hierarchy and discuss its importance in computer organization. | L2 | CO4 | [10M] |
| | **b.** | Explain auxiliary memory with characteristics and applications. | L2 | CO4 | [10M] |
| | | **(OR)** | | | |
| **8** | **a.** | Explain the concept of virtual memory and paging mechanism. | L2 | CO4 | [10M] |
| | **b.** | Interpret priority interrupt system and DMA controller operation. | L2 | CO4 | [10M] |
| **UNIT-V** | | | | | |

MODEL QUESTION PAPER

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| **9** | **a.** | Explain the characteristics of multiprocessor systems. | L2 | CO5 | [10M] |
| | **b.** | Discuss interprocessor communication and synchronization techniques. | L3 | CO5 | [10M] |
| **(OR)** | | | | | |
| **10** | **a.** | Demonstrate instruction pipelining with timing diagram. | L2 | CO5 | [10M] |
| | **b.** | Explain RISC pipeline architecture and delayed branch concept. | L2 | CO5 | [10M] |

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MODEL QUESTION PAPER